SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech (III Year / II Sem)	**Academic Year:** 2025–2026
Course Title: AI Assisted Coding (23CS002PC304)	**Regulation:** R23 | **Lab Session:** Week 11 – Monday
Assignment No: Lab 22 (Advanced Ethical Considerations)	**Assignment Type:** Ethics & Safety Lab Report

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LAB 22: ADVANCED ETHICAL CONSIDERATIONS – ADDRESSING COMPLEX SCENARIOS AND LIMITATIONS

Lab Objectives:

- Understand ethical concerns in AI-assisted coding and algorithmic gatekeeping.
- Explore complex real-world scenarios where AI decisions have life-and-death consequences.
- Learn to identify limitations and societal biases in AI-generated solutions.
- Encourage critical thinking and human professional responsibility in AI collaboration.

Task 1 – Bias Detection in AI Suggestions

Experiment: We tested three prompt variations for applicant ranking algorithms using identical candidate competencies.

Candidate Profile	Test Score	College Tier	Career Break Context	Neutral Score	Biased AI Score	Disparity
Candidate 1: Ananya Sharma	96%	Tier 2	Maternity / Caregiving	**94.0**	**29.8**	**-64.2 pts**
Candidate 2: Rohan Verma	74%	Tier 1	None (Unbroken)	**61.0**	**59.2**	**-1.8 pts**
Candidate 3: D. Poornanda Chalam	92%	Tier 2	None (Unbroken)	**88.0**	**44.6**	**-43.4 pts**

Ethical Discussion & Findings:

1. **Hidden Assumptions in AI Responses:** The AI model codified a linear tenure bias, assuming unbroken employment is the sole measure of dedication. It penalized caregiving breaks by 20 points while rewarding institutional brand pedigree by 25 points.
2. **Impact on Fairness & Accessibility:** Automated ATS filtering tools silently discard qualified female engineers and rural graduates, homogenizing engineering teams and perpetuating systemic socioeconomic divides.

Task 2 – Privacy and Sensitive Data Handling

Demonstrated Vulnerability: AI scripts routinely log raw PII (Names, Phone Numbers, Aadhaar IDs, Medical Diagnoses) into unencrypted log files.

Anonymization Pipeline Implemented:

- **Masked Phone Numbers:** `+91-9876543210` → `+91-******3210`
- **Masked Aadhaar Numbers:** `4521-8890-1234` → `XXXX-XXXX-1234`

- **Salted Pseudonymization:** User ID → Irreversible hash token `ANON_35DCC96884B4`
- **K-Anonymity Generalization:** Raw Salary → Salary Bracket (`INR 5 - 10 LPA`)

Legal Frameworks & Developer Duties:

- **Digital Personal Data Protection (DPDP) Act 2023 (India):** Imposes penalties up to **INR 250 Crores** for failure to implement reasonable security safeguards.
- **GDPR (EU Article 83):** Fines up to **20 Million Euros or 4% of worldwide turnover**.
- **Developer Responsibility:** Under "Privacy by Design", developers are legal fiduciaries. Ignorance of AI behavior is not a valid legal defense.

Task 3 – AI-Generated Security Risks & Safe Usage Guidelines

AI tools frequently output obsolete cryptographic algorithms (MD5, plain SHA-256) and hardcoded secrets.

Hardened Production Authentication:

```
class ProductionAuthSystem: ITERATIONS = 600000 # OWASP recommended minimum for PBKDF2-HMAC-SHA256
@classmethod def hash_password_secure(cls, password: str): salt = secrets.token_hex(16) key =
hashlib.pbkdf2_hmac( hash_name="sha256", password=password.encode("utf-8"), salt=salt.encode("utf-8"),
iterations=cls.ITERATIONS, ) return key.hex(), salt @classmethod def verify_password_secure(cls, password:
str, salt: str, expected_hash: str): computed_key = hashlib.pbkdf2_hmac( hash_name="sha256",
password=password.encode("utf-8"), salt=salt.encode("utf-8"), iterations=cls.ITERATIONS, ) return
secrets.compare_digest(computed_key.hex(), expected_hash)
```

4 Ethical Guidelines for Safe AI Code Usage:

1. **Zero-Trust Verification:** Treat all AI-generated code as untrusted user input until audited.
2. **Mandatory Static Analysis (SAST):** Enforce Bandit/SonarQube scans on all AI pull requests.
3. **Cryptographic Hygiene:** Ban deprecated algorithms (MD5, SHA1, DES) across repositories.
4. **Secrets Decoupling:** Prohibit hardcoded tokens; mandate secrets vault managers.

Task 4 – Accountability in Human–AI Collaboration (Hospital ICU Case)

> **Scenario:** AI writes automated drug dosage calculation logic for an ICU patient monitor. Due to an unhandled sensor desync, the system delays an adrenaline injection, injuring the patient.

Stakeholder	Role in Failure	Legal & Moral Liability
AI Tool Vendor	Provided general probabilistic LLM	Protected by Terms of Service ("as-is", non-liability clauses). Models are not certified medical diagnostic software.
Human Developer	Merged code without edge-case testing	**Primary Professional Liability.** Engineers hold a professional duty of care. Merging unverified code constitutes gross negligence.
Hospital Trust	Deployed automated actuation without oversight	**Institutional & Vicarious Liability.** Failure of clinical governance and failure to enforce human-in-the-loop nursing protocols.

```
STUDENT REFLECTION REPORT SUMMARY (D. POORNANDA CHALAM):
1. AI is an instrument, not an agent: Blaming an AI algorithm is equivalent to an architect blaming a
calculator for a collapsed bridge.
2. The developer who signs off on the release assumes 100% engineering ownership.
3. In life-critical contexts, autonomous actuation must be gated by mandatory clinical validation and fail-
safe analog interlocks.
```

Task 5 – Limitations of AI in Critical Systems (Railway Case)

AI code generated for railway interlocking used naive Boolean logic: `if train_approaching and track_clear: return "GREEN"`.

Why Direct AI Code Fails in Safety-Critical Domains (CENELEC EN 50128 SIL-4 / DO-178C):

- **Probabilistic Black Box:** Safety-critical systems require formal deterministic state machines and mathematical proofs. LLMs cannot guarantee determinism.
- **Missing Fail-Safe Defaults:** If sensor communication drops or power fluctuates, the system must physically drop to a safe state (**Fail-Closed / Gravity Drop to RED**).
- **Dual Sensor Redundancy:** Must enforce 2-out-of-2 voting and mechanical switch interlocking before any signal clears.

Balancing Efficiency vs Safety:

Restrict AI to auxiliary developer workflows (generating documentation, formatting test cases, synthesizing fuzzing data). **All core control logic, safety interlocks, and emergency overrides must remain 100% authored and verified by certified human engineers.**

Task 6 – Ethical Use of AI-Generated Code: Case Studies

Real-World Case Study	Core Controversy	Student Engineering Takeaway
Doe v. GitHub / Microsoft / OpenAI	Training on public code without attribution (DMCA 1202).	Must verify AI outputs do not emit verbatim copyrighted proprietary blocks.
GPL "Copyleft Contamination"	AI outputting GPL-licensed code into closed commercial repos.	Software Composition Analysis (SCA) tools are mandatory to avoid forced open-sourcing.
Samsung Semiconductor Leak (2023)	Engineers pasted confidential chip code into public AI chats.	Mandate private enterprise AI endpoints; ban pasting proprietary IP into public LLMs.

The Three Pillars for Ethically Using AI Code:

1. **ATTRIBUTION:** Maintain an AI collaboration ledger in git commit history and verify algorithm licenses with public search engines.
2. **VERIFICATION:** Require 100% branch test coverage, static security analysis, and whiteboard peer reviews.
3. **RESPONSIBLE DEPLOYMENT:** Implement canary releases, automated rollback triggers, and assign named human engineers as accountable owners.

LAB 22 COMPLETE | ALL 6 TASKS VERIFIED IN VS CODE TERMINAL

D. POORNANDA CHALAM (2503A52361) | BATCH 13